Mid-term Examination (Morning)

Model Answers

SoLo - Level 2

Q1 (a) $x(t) = 3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t)$

$f_s = 5000$ Hz

$f_1 = 1000$ Hz ; $f_2 = 3000$ Hz ; $f_3 = 6000$ Hz

($f_1 < \frac{f_s}{2}$ not affected)

$f_2 > \frac{f_s}{2}$ so it is affected by aliasing

$f_2' = f_2 - k \cdot f_s$ (k = 1)

$= 3000 - 5000$

$= -2000$ Hz

$f_3 > \frac{f_s}{2}$ affected by aliasing

$f_3' = f_3 - 1 \cdot 5000$

$= 6000 - 5000$

$= 1000$ Hz

[2]

After sampling there will be two frequencies ($f_1 = f_3'$ & $f_2 \to f_2'$)

$x[n] = 3\cos\left(2\pi \frac{1000}{5000} n\right) + 5\sin\left(2\pi \left(\frac{-2000}{5000}\right) n\right) + 10\cos\left(2\pi \frac{1000}{5000} n\right)$

[2] $\boxed{x[n] = 13\cos\left[\frac{2\pi}{5} n\right] - 5\sin\left[\frac{4\pi}{5} n\right]}$

(b) $f_s = 2(10\text{kHz} - 8\text{kHz}) = 4\text{k Hz}$ $T = \frac{1}{f_s}$ $f_1 = 8\text{kHz}, f_2 = 10\text{kHz}$

aliased frequency $f_0' = f_0 - k f_s$, $k = \pm 1, \pm 2, \pm 3 \cdots$

Let $k = 2$

$f_1' = 8 - 2 \times 4 = 0$

$f_2' = 10 - 2 \times 4 = 2\text{kHz}$ } (A)

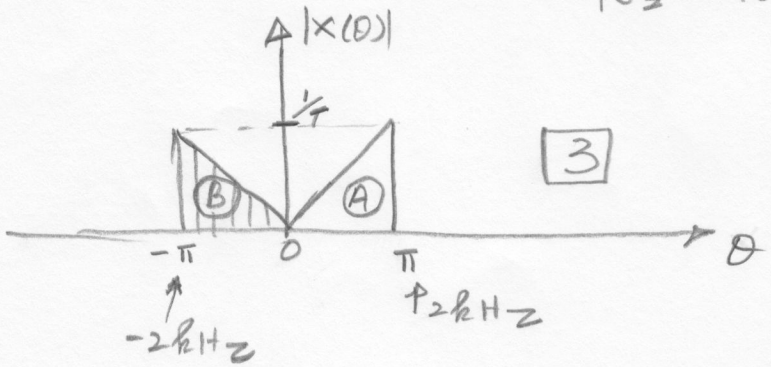
[1]

Let $k = -2$

$f_1' = -8 - (-2) \times 4 = 0$

$f_2' = -10 - (-2) \times 4 = -2$ } (B)

[1]

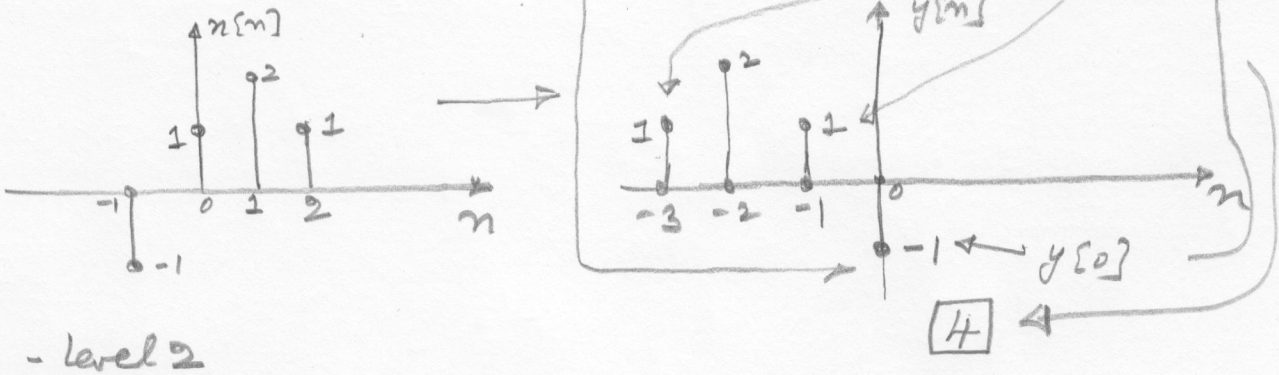


[3]

$Q_1(c)$ $y[n] = x[-n-1]$

There are many methods to do this questions.
The simple one is:

Let $n=0$ $y[0] = x[-1]$ → $n=-1$ $y[-1] = x[0]$
$n=1$ $y[1] = x[-2]$ → $n=-2$ $y[-2] = x[1]$
$n=2$ $y[2] = x[-3]$ → $n=-3$ $y[-3] = x[2]$



4

SOLO - level 2

$Q2(a)$ (i) $H(z) = \frac{k}{1-az^{-1}}$ $0<a<1$

This is a Low pass filter & the maximum value occurs at $\theta = 0$

$$H(\theta)\Big|_{\theta=0} = 1 = \frac{k}{1-a} \quad \therefore \quad k = (1-a)$$ 2

(ii) $$H(\theta) = \frac{1-a}{1-a e^{-j\theta}}$$

$$\therefore |H(\theta)|^2 = H(\theta)\cdot H^*(\theta) = \frac{(1-a)}{1-a e^{-j\theta}} \cdot \frac{1-a}{1-a e^{j\theta}}$$

2 $$|H(\theta)|^2 = \frac{(1-a)^2}{1+a^2-a(e^{j\theta}-e^{-j\theta})} = \frac{(1-a)^2}{1+a^2-2a\cos\theta}$$

at $\theta = \theta_c$, $|H(\theta)|^2 = \frac{1}{2}$ (3dB cut-off)

2 $$\therefore \frac{1}{2} = \frac{(1-a)^2}{1+a^2-2a\cos\theta_c}$$

$$1+a^2-2a\cos\theta_c = 2-4a+2a^2$$

$$a^2 - 2a(2-\cos\theta_c) + 1 = 0$$

$$\therefore a = \frac{2a(2-\cos\theta_c) \pm \sqrt{4a^2(2-\cos\theta_c)^2 - 4\cdot a^2}}{2a}$$

$$a = (2-\cos\theta_c) \pm \sqrt{(2-\cos\theta_c)^2 - 1} \Rightarrow (2-\cos\theta_c)^2 - 1 \geq 0$$

1 $$\therefore a = (2-\cos\theta_c) - \sqrt{(2-\cos\theta_c)^2-1}$$ $2-\cos\theta_c \geq \pm 1$

$a<1$ [given]

Q2 (b)

$$H(z) = \frac{\frac{1}{2}}{1-\frac{1}{2}z^{-1}} + \boxed{\frac{1}{(1-\frac{1}{2}z^{-1})(1-z^{-1})}}$$

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}z^{-1}} + \boxed{\frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1-z^{-1}}}$$

Use any methods to find A & B

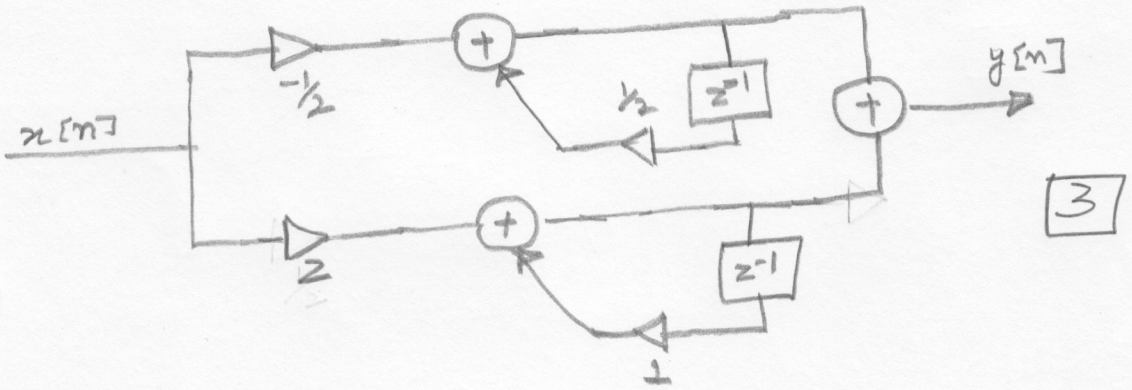
$A = -1$

$B = 2$ A

[2]

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}z^{-1}} - \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{2}{1-z^{-1}}$$

$$= \frac{-\frac{1}{2}}{1-\frac{1}{2}z^{-1}} + \frac{2}{1-z^{-1}} \leftarrow [1]$$



[3]

SOLO-Level 2

Q3 (a) $y[n] = a_0 x[n] + \frac{1}{3} x[n-1] + a_2 x[n-2]$

$$Y(z) = a_0 X(z) + \frac{1}{3} X(z) z^{-1} + a_2 X(z) z^{-2}$$

$$H(z) = a_0 + \frac{1}{3}z^{-1} + a_2 z^{-2}$$

$$H(\theta)\Big|_{\theta=0} = 1 = a_0 + \frac{1}{3} + a_2 \qquad [2]$$

For linear phase $a_0 = a_2$ (symmetrical coefficients)

$$\therefore 1 = a_0 + \frac{1}{3} + a_0$$

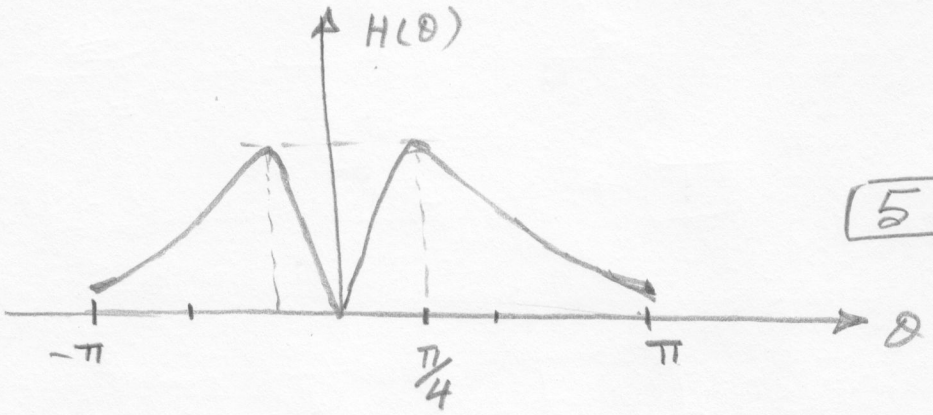
$$\therefore a_0 = \frac{1}{3}$$

$$\boxed{\therefore a_0 = \frac{1}{3} \ \& \ a_2 = \frac{1}{3}} \qquad [2]$$

$Q_3(b)$ $\quad H(z) = \dfrac{z(z-1)}{\left(z-\frac{1}{2}-\frac{1}{2}j\right)\left(z-\frac{1}{2}+\frac{1}{2}j\right)}$ [2]

$= \dfrac{z(z-1)}{z^2-z+\frac{1}{2}}$ [2]

(c)



[5]

SOLO - Level 2

$Q_4(a)$

[2]

$Y(z) = X(z) - X(z)z^{-1} \Rightarrow H(z) = 1 - z^{-1}$

$H(\theta) = 1 - e^{-j\theta} = e^{-j\frac{\theta}{2}}\left[e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}\right]$

$= e^{-j\frac{\theta}{2}}\left[\dfrac{e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}}{2j}\right] \times 2j$ ($j = e^{j\pi/2}$)

$= e^{-j\frac{\theta}{2}} \cdot e^{j\frac{\pi}{2}}$ $\quad 2\sin\frac{\theta}{2}$ = $B(\theta)$

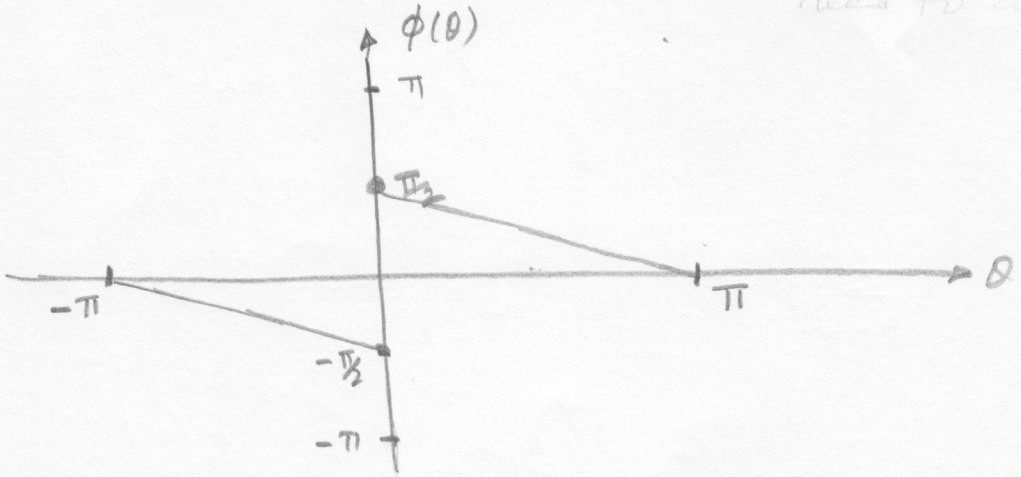
$= e^{j(\frac{\pi}{2}-\frac{\theta}{2})}$

$\phi(\theta) = \frac{\pi}{2} - \frac{\theta}{2}$ ← $B(\theta) \geq 0$

$= \frac{\pi}{2} - \frac{\theta}{2} - \pi = -\frac{\pi}{2} - \frac{\theta}{2}$ ← when $B(\theta) < 0 \Rightarrow -1 = e^{-j\pi}$

need to edit

[2]



Q4 (b) (i) 4-point DFT $\Rightarrow h[n] = \{1, 2, 3, 0\}$ include zero [2]

$$H(k) = \sum_{n=0}^{3} h[n]\, e^{-j\frac{2\pi n k}{4}} \quad [1]$$

(ii) If $k = 3$, then

$$H(3) = \sum_{n=0}^{3} h(n)\, e^{-j\frac{2\pi n \times 3}{4}}$$

$$= 1 + 2e^{-j\frac{3\pi}{2}} + 3e^{-j3\pi} + 0$$

$$= 1 + 2j + (-3) + 0$$

$$H(3) = -2 + 2j \quad [2]$$

(iii) DC gain is $H(0)$

$$\therefore H(0) = 1 + 2\times1 + 3\times1 + 0\times1 = 6 \quad [2]$$

(iv) Using conjugate symmetry property

$$H(1) = H(3)^{*} = -2 - 2j \quad [2]$$

SOLO - Level 3

(a) (i) $h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta)\, e^{j\theta n}\, d\theta = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} e^{-j3\theta} e^{j\theta n}\, d\theta$ — 2

$= \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} e^{j(n-3)\theta}\, d\theta = \frac{1}{2\pi}\left[\frac{e^{j(n-3)\theta}}{j(n-3)}\right]_{-\pi/2}^{\pi/2}$ — 3

$= \frac{1}{2\pi j(n-3)}\left(e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2}\right)$

$= \frac{1}{\pi(n-3)}$

$\frac{e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2}}{2j} = \sin(n-3)\frac{\pi}{2}$

$= \frac{1}{\pi}\,\frac{\sin(n-3)\frac{\pi}{2}}{(n-3)}$ ← $n \neq 3$ — 2

(ii) For $n = 3$ $\quad h_d[3] = 1/2$

(iii) $n=0 \quad h_d[0] = \frac{\sin(-3\pi/2)}{(-3\pi)} = \frac{-1}{3\pi}$

$n=1 \quad h_d[1] = \frac{\sin(-2\pi/2)}{-2\pi} = 0$

$n=2 \quad h_d[2] = \frac{\sin(-\pi/2)}{-\pi} = \frac{1}{\pi}$

$n=3 \quad h_d[3] = 1/2$ — 3

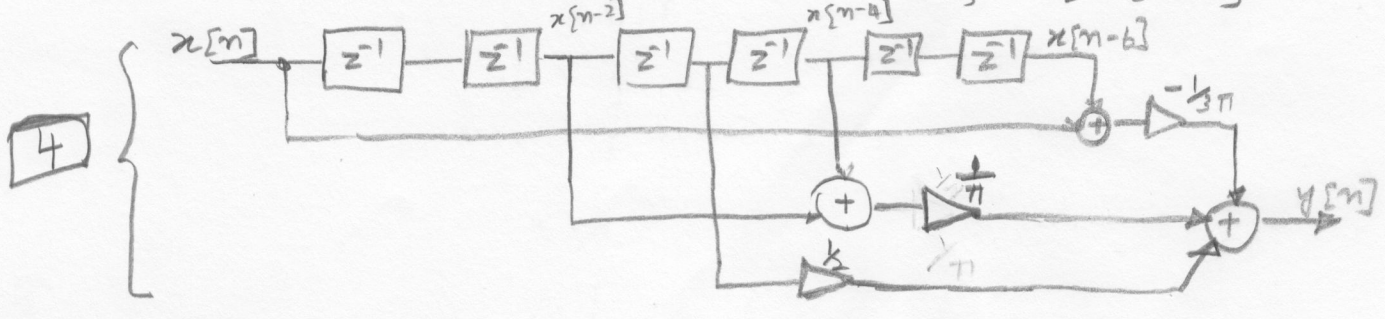
(iii)

$h_d[n]$	$-\frac{1}{3\pi}$	0	$\frac{1}{\pi}$	$\frac{1}{2}$	$\frac{1}{\pi}$	0	$-\frac{1}{3\pi}$
n	0	1	2	3	4	5	6

3

linear phase filter → coefficients are symmetrical

(iv) $y[n] = -\frac{1}{3\pi}\{x[n] + x[n-6]\} + \frac{1}{\pi}\{x[n-2] + x[n-4]\} + \frac{1}{2}x[n-3]$



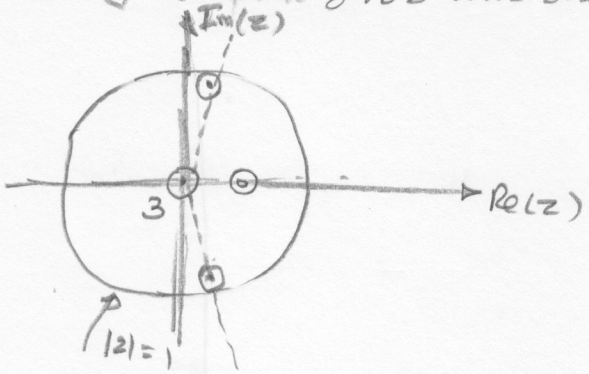
4

(b)

$$H(z) = (1 - 0.3z^{-1} + 0.9z^{-2})(1 - 0.5z^{-1})$$

This filter has a pair of complex zeros and one real zero.

* The magnitude response of this filter will not be changed if we flip the zeros to their reciprocal location.



The three other FIR filters that have the same magnitude responses are

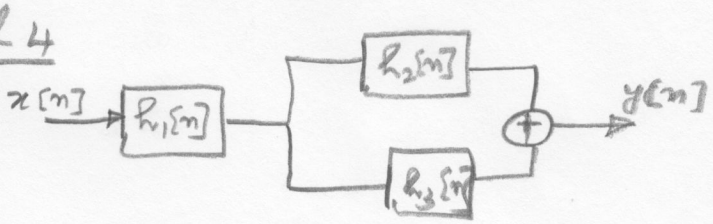
[2] $H_1(z) = (1 - 0.3z^{-1} + 0.9z^{-2})(0.5 - z^{-1})$ ← Reflect real zero

[2] $H_2(z) = (0.9 - 0.3z^{-1} + z^{-2})(1 - 0.5z^{-1})$ ← Reflect complex zeros

[2] $H_3(z) = (0.9 - 0.3z^{-1} + z^{-2})(0.5 - z^{-1})$ ← Reflect complex zeros & real zero.

Note that each of these filters is causal with $h[n] = 0$ for $n > 3$

SOLO - Level 4



$h_1[n] = u[n-2]$
$h_2[n] = n\,u[n]$
$h_3[n] = \delta[n-2]$

$y[n] = u[n-2] * \delta[n-2] + u[n-2] * n\,u[n]$ ← [2]

[2] $= \sum_{k=-\infty}^{\infty} u[k-2]\,\delta[n-k-2] + \sum_{k=-\infty}^{\infty} u[k-2]\cdot(n-k)\,u[n-k]$

$k-2 \geq 0$, $k \geq 2$; $n-k-2 = 0$, $\therefore k = n-2$ — Two limits of the summation

$k-2 \geq 0$, $k \geq 2$; $n-k \geq 0$, $k \leq n$ — Two limits of the summation

[1] $= \sum_{k=2}^{n-2} \delta[n-k-2] + \sum_{2}^{n} n-k$

$= \delta[n-4] + \delta[n-5] + \delta[n-6] + \cdots + \delta[0]$

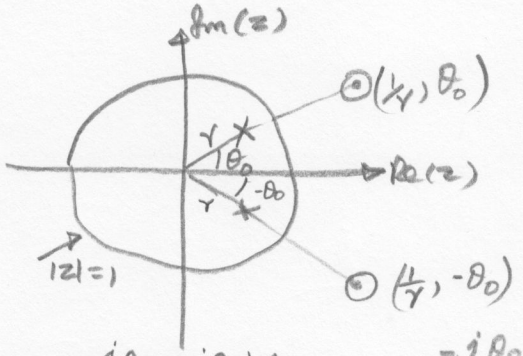
$= u[n-4] + (n-2) + (n-3) + \cdots + 1 + 0$

arithmetic sum of $(n-2)$ terms ($(n-1)$ terms, $d=1$, $a=0$)

$= u[n-4] + \frac{n-2}{2}\{2(1) + ((n-2)-1)1\}$ $\quad S = \frac{n}{2}\{2a + (n-1)d\}$

[3] $y[n] = u[n-4] + \frac{1}{2}(n-2)(n-1)\,u[n-2]$ ← [1]

(b)



$$H(z)=\frac{\left(1-\frac{1}{r}e^{j\theta_0}z^{-1}\right)\left(1-\frac{1}{r}e^{-j\theta_0}z^{-1}\right)}{\left(1-re^{j\theta_0}z^{-1}\right)\left(1-re^{-j\theta_0}z^{-1}\right)}$$

3

6

$$H(\theta)=\frac{\left(1-\frac{1}{r}e^{j\theta_0}\cdot e^{-j\theta}\right)\left(1-\frac{1}{r}e^{-j\theta_0}\cdot e^{-j\theta}\right)}{\left(1-re^{j\theta_0}e^{-j\theta}\right)\left(1-re^{-j\theta_0}e^{-j\theta}\right)}$$

$$=\frac{1}{r^2}\,\frac{\left(r-e^{j\theta_0}e^{-j\theta}\right)\left(r-e^{-j\theta_0}\cdot e^{-j\theta}\right)}{\left(1-re^{j\theta_0}e^{-j\theta}\right)\left(1-re^{-j\theta_0}\cdot e^{-j\theta}\right)}$$

$$H(\theta)=\frac{1}{r^2}\,\frac{\left\{r-\cos(\theta_0-\theta)-j\sin(\theta_0-\theta)\right\}}{\left\{1-r\cos(\theta_0-\theta)-jr\sin(\theta_0-\theta)\right\}}\cdot\frac{\left\{r-\cos(\theta_0+\theta)+j\sin(\theta_0+\theta)\right\}}{\left\{1-r\cos(\theta_0+\theta)+jr\sin(\theta_0+\theta)\right\}}$$

Modulus of the numerator will cancel the modulus of the numerator

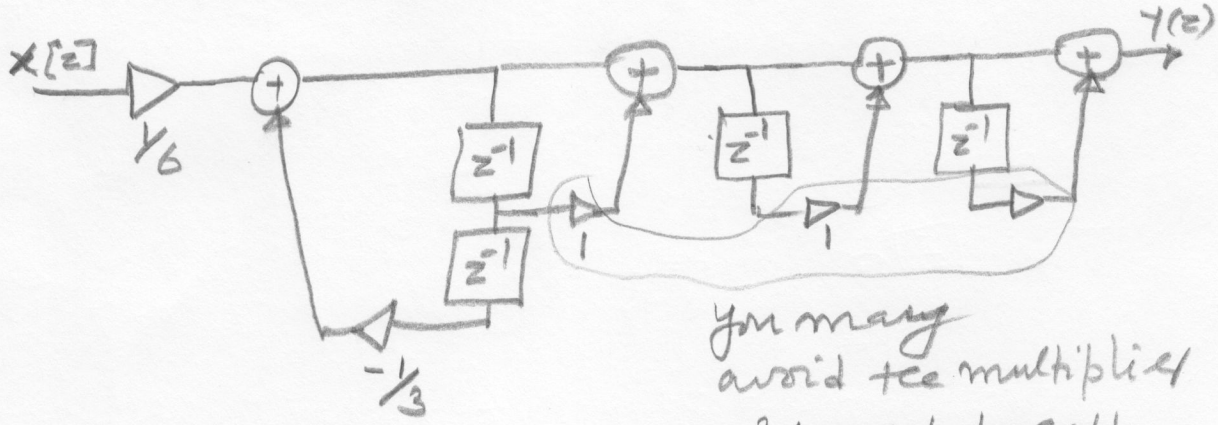
$\therefore |H(\theta)|=\frac{1}{r^2}$ is an all-pass filter

(c)

$$G(z)=\frac{(z+1)^3}{6z\left(z-j\frac{1}{\sqrt{3}}\right)\left(z+j\frac{1}{\sqrt{3}}\right)}=\frac{(z+1)^3}{6z\left(z^2+\frac{1}{3}\right)}$$

$$G(z)=\frac{(1+z^{-1})^3}{6\left(1+\frac{1}{3}z^{-2}\right)}=\frac{1}{6}\,\frac{(1+z^{-1})(1+z^{-1})(1+z^{-1})}{\left(1+\frac{1}{3}z^{-2}\right)}$$

2



you may avoid the multiplier & connect directly

3

SOLO - Level 5

Because $h[n]=0$ for $n<0$ & for $n>7$, $H(z)$ has seven zeros. With a complex zero at $z=0.4\,e^{j\frac{\pi}{3}}$, because $h[n]$ is real, there must be another zero at the conjugate location $z=0.4\,e^{-j\frac{\pi}{3}}$.

[2] $$H_1(z)=(1-0.4e^{j\frac{\pi}{3}}z^{-1})(1-0.4e^{-j\frac{\pi}{3}}z^{-1})=1-0.4z^{-1}+0.16z^{-2}$$

The linear phase constraint requires that there be a pair of zeros at their reciprocal locations. Therefor $H(z)$ must also contain the factor

[3] $$H_2(z)=0.16-0.4z^{-1}+z^{-2}$$

The system function also contains a zero at $z=3$ again, for linear phase, there should be a zero at $z=\frac{1}{3}$. Thus $H(z)$ also has the factor

[2] $$H_3(z)=(1-3z^{-1})(1-\tfrac{1}{3}z^{-1})$$

We have

$$H(z)=k\,(1-0.4z^{-1}+0.16z^{-2})(0.16-0.4z^{-1}+z^{-2})(1-3z^{-1})(1-\tfrac{1}{3}z^{-1})$$

[3] We have $h[0]=1$ ← given

∴ The zero-order term in $H(z)$ is $0.16k$,

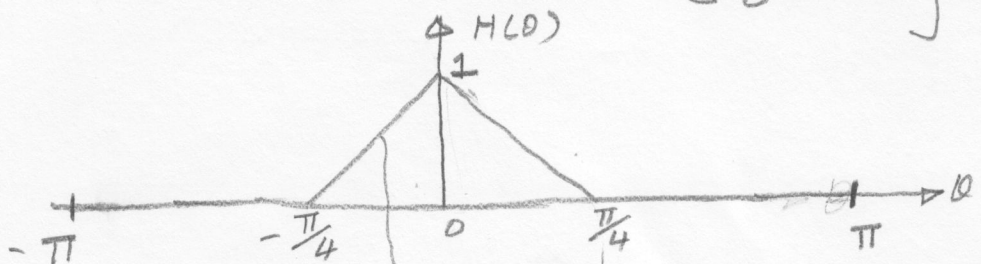
$\therefore 0.16k=1$ $\quad[h[0]=1]$

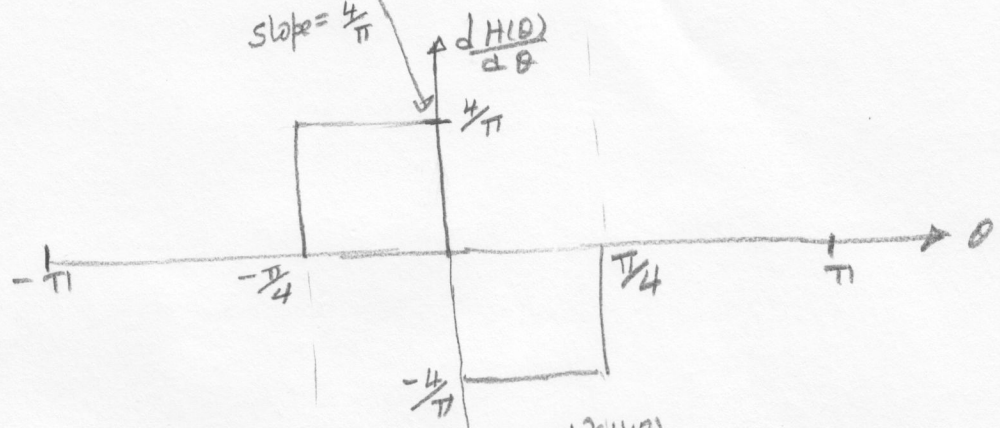
$k=\frac{1}{0.16}$

$$H(z)=\frac{1}{0.16}(\quad)(\quad)(\quad)(\quad)$$

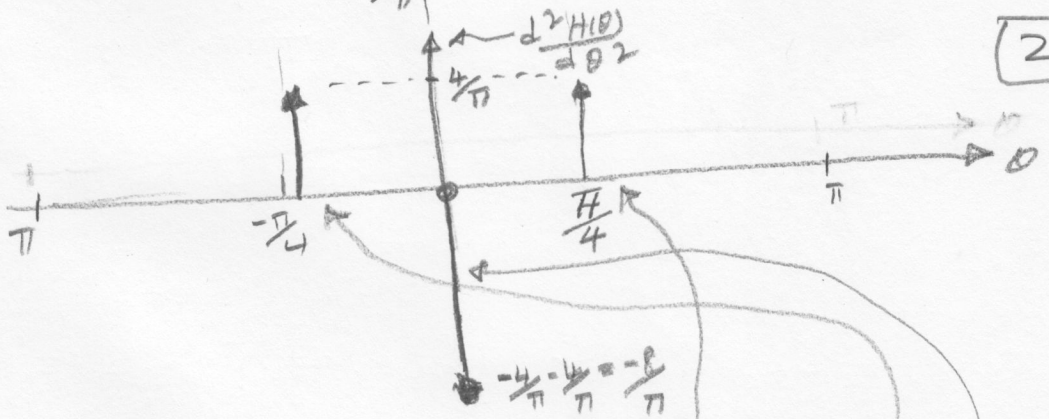
(b) To find the impulse response, we take IDTFT of $X(\theta)$. We can use differentiation property of DTFT

$$n h[n] \overset{DTFT}{\Longleftrightarrow} j \frac{d H(\theta)}{d\theta} \quad \} \quad \boxed{1}$$





$\boxed{2}$



$\boxed{2}$

$\boxed{2}$ Taking IDTFT of $\frac{d^2H(\theta)}{d\theta^2} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{d^2H(\theta)}{d\theta^2} e^{jn\theta} d\theta$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left\{ \frac{4}{\pi}\left[\delta(\theta-\tfrac{\pi}{4}) + \delta(\theta+\tfrac{\pi}{4})\right] - \frac{8}{\pi}\delta[0] \right\} e^{jn\theta} d\theta$$

$\boxed{1}$

$$= \frac{1}{2\pi}\left\{ \frac{4}{\pi}\left(e^{jn\frac{\pi}{4}} + e^{-jn\frac{\pi}{4}}\right) - \frac{8}{\pi}\left(\frac{e^{jn\pi} - e^{-jn\pi}}{jn}\right)\right\} = \frac{4}{\pi^2}\cos\frac{n\pi}{4} - \frac{8}{\pi}\frac{\sin n\pi}{n\pi}$$

$\frac{\sin n\pi}{n\pi} = 1$

$$= \frac{4}{\pi^2}\cos\frac{n\pi}{4} - \frac{8}{\pi}$$

$\boxed{2}$ To get $h[n]$ multiply by $\frac{j}{n}\cdot\frac{j}{n} = \frac{-1}{n^2}$ $\left\{ n h[n] \Longleftrightarrow j\frac{dH(\theta)}{d\theta} \right.$

$$\boxed{h[n] = \frac{8}{\pi n^2} - \frac{4}{\pi^2 n^2}\cos\frac{n\pi}{4}}$$

(c) $|X(\theta)|^2 = 1+a^2-2a\cos\theta \qquad 0<a<1 \qquad x[0]>0$

3 pts $\quad |X(\theta)|^2 = 1+a^2-2a\,\frac{e^{j\theta}+e^{-j\theta}}{2} = 1+a^2-2e^{j\theta}-2e^{-j\theta}$

Replacing $e^{j\theta}$ with z & $e^{-j\theta}$ with z^{-1}, we have

$$G(z) = \overbrace{X(z)\cdot X(z^{-1})}^{=|H(\theta)|^2} = 1+a^2-az-az^{-1} \leftarrow [3]$$

$$= \underbrace{(1-az^{-1})}_{\text{causal}}\underbrace{(1-az^{1})}_{\text{noncausal}} \leftarrow [1]$$

Therefore, a real valued causal sequence with the given magnitude $x[0]>0$ is

$$\boxed{x[n] = \delta[n] - a\,\delta[n-1]} \leftarrow [2]$$

(c) $|X(\theta)|^2 = 1+a^2-2a\cos\theta \qquad 0<a<1 \qquad x[0]>0$

3 pts $\quad |X(\theta)|^2 = 1+a^2-2a\,\frac{e^{j\theta}+e^{-j\theta}}{2} = 1+a^2-2e^{j\theta}-2e^{-j\theta}$

Replacing $e^{j\theta}$ with z & $e^{-j\theta}$ with z^{-1}, we have

$$G(z) = \overbrace{X(z)\cdot X(z^{-1})}^{=|H(\theta)|^2} = 1+a^2-az-az^{-1} \leftarrow [3]$$

$$= \underbrace{(1-az^{-1})}_{\text{causal}}\underbrace{(1-az^{1})}_{\text{noncausal}} \leftarrow [1]$$

Therefore, a real valued causal sequence with the given magnitude $x[0]>0$ is

$$\boxed{x[n] = \delta[n] - a\,\delta[n-1]} \leftarrow [2]$$